

Anya Agarwal

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Education

University of Pennsylvania School of Engineering & Applied Science

December 2026

M.S.E in Computer Graphics and Game Technology

University of California, Berkeley College of Computing, Data Science, and Society

May 2025

B.A. in Computer Science, Certificate in Design Innovation

GPA: 3.86

Relevant Courses: GPU Programming, Computer Graphics, Computer Animation, 3D Computer Modeling, Data Structures, Algorithms, Signals and Systems, Machine Learning, Operating Systems, Computer Architecture, User Interfaces, Probability, Data Science, Prototyping and Fabrication, Design Methodology

Skills

Languages: CUDA, C, C++, WebGPU, Python, Java, React.js, HTML, Javascript, Typescript, Tailwind CSS, Rust, Pandas, SQL

Tools: Autodesk Maya, Unity, Adobe Photoshop & After Effects & Premiere Pro, Figma

Projects

CUDA GPU Pathtracer

October 2025

- Built a GPU-accelerated pathtracer using CUDA and C++, supporting multiple material shaders, antialiasing, and more.
- Integrated bounding volume hierarchy acceleration to optimize loading .OBJ files, creating a 9x speedup for large meshes.
- Added the ability to make fast and complex renders with texture mapping, and stream compaction for path termination.

CUDA Boids Flocking Simulator

September 2025

- Developed a real-time, optimized flocking simulation on the GPU using CUDA, based on the Reynolds Boids algorithm.
- Implemented spatial-partitioning algorithms for memory coherence, resulting in a speedup from quadratic to linear time.
- Used CUDA event timers and NVIDIA NSight platforms to analyze and evaluate performance of flocking simulations.

Cloth Simulator

April 2024

- Utilized mass and spring system to create a cloth simulation and its corresponding collisions with other objects.
- Created Blinn-Phong, bump, displacement, texture, and other shaders for the cloth using the OpenGL API.

Experience

Berkeley Institute of Design

September 2024 - May 2025

Undergraduate Researcher

- Assisted in conducting and writing a research paper about the creation of Noise Pilot — a tool that implements diffusion-based image generation in a node-based image processing interface — and its impact and usability for artists.
- Connected Noise Pilot with a drawing interface API to allow users to draw masks and shapes for their image generation
- Implemented an inverse kinematics algorithm from a research paper to explore movement in VR environments.

LearnQ.ai

June 2023 - August 2023

Fullstack Software Engineer Intern

- Implemented a complete UI for new AI chatbot feature and menu page using React Typescript and Tailwind CSS.
- Generated several AI chatbot modes using LangChain and OpenAI, and trained them on varying knowledge bases.
- Designed a framework for the chatbot to handle connections between frontend and backend modules.

Honors/Awards

Grace Hopper Celebration Student Scholarship

June 2023

- Scholarship for women and non-binary undergraduates to attend the Grace Hopper Celebration and participate in various networking and skill building workshops.